

NAME:

NO:

CLASS:

RIVERSIDE SECONDARY SCHOOL



PRELIMINARY EXAMINATION 2026

SUBJECT : ADDITIONAL MATHEMATICS

SUBJECT CODE /PAPER : 4049/02

LEVEL/STREAM : Sec 4 Express

DURATION : 2 hours 15 minutes

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your class, index number and name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures or in 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an electronic calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

At the end of the examination fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

The total of the marks for this paper is 90.

This document consists of **21** printed pages and **1** blank page.

[Turn Over

Mathematical Formulae**1. ALGEBRA***Quadratic Equation*

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Binomial expansion

$$(a + b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + \binom{n}{r}a^{n-r}b^r + \dots + b^n,$$

where n is positive integer and $\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)\dots(n-r+1)}{r!}$

2. TRIGONOMETRY*Identities*

$$\sin^2 A + \cos^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

Formulae for $\triangle ABC$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\Delta = \frac{1}{2} bc \sin A$$

Answer all the questions.

- 1 Solve the equation $\sqrt{2x+7} - \sqrt{x+3} = 1$. [5]

$$\left(\sqrt{2x+7}\right)^2 = \left(1 + \sqrt{x+3}\right)^2$$

$$2x+7 = 1 + 2\sqrt{x+3} + (x+3)$$

$$x+3 = 2\sqrt{x+3}$$

$$(x+3)^2 = 4(x+3)$$

$$(x+3)(x-1) = 0$$

$$x = -3 \text{ or } x = 1$$



- 2 In the expansion of $\left(1 + \frac{x}{2}\right)^n$, where n is a positive integer, the ratio of the coefficient of x^3 to the coefficient of x^2 is $7:6$.
- (a) Find the value of n . [4]

$$\frac{\binom{n}{3}\left(\frac{1}{2}\right)^3}{\binom{n}{2}\left(\frac{1}{2}\right)^2} = \frac{7}{6}$$

$$7\binom{n}{2}\left(\frac{1}{2}\right)^2 = 6\binom{n}{3}\left(\frac{1}{2}\right)^3$$

$$7\left(\frac{n(n-1)}{2}\right)\left(\frac{1}{4}\right) = 6\left(\frac{n(n-1)(n-2)}{6}\right)\left(\frac{1}{8}\right)$$

$$\frac{7}{8}n(n-1) = \frac{1}{8}n(n-1)(n-2)$$

$$n(n-1)\left[\frac{7}{8} - \frac{1}{8}(n-2)\right] = 0$$

$$\frac{7}{8} = \frac{1}{8}(n-2) \quad n(n-1) = 0 \text{ (rej.)}$$

$$n-2 = 7$$

$$n = 9$$

(b) Hence find the coefficient of x^2 in the expansion of $(3-2x)^2 \left(1+\frac{x}{2}\right)^9$. [3]

$$(3-2x)^2 = 9 - 12x + 4x^2$$

$$\left(1+\frac{x}{2}\right)^9 = 1 + \frac{9}{2}x + 9x^2 + \dots$$

$$\text{Coefficient of } x^2: 9(9) + (-12)\left(\frac{9}{2}\right) + 4(1)$$

$$= 31$$

- 3 (a) Solve the equation $2\log_4 x - \log_x 4 = 1$.

[4]

$$2\log_4 x - \frac{1}{\log_4 x} = 1$$

Let $u = \log_4 x$:

$$2u - \frac{1}{u} = 1$$

$$2u^2 - u - 1 = 0$$

$$(2u + 1)(u - 1) = 0$$

$$u = -\frac{1}{2} \text{ or } u = 1$$

$$\log_4 x = -\frac{1}{2} \text{ or } 1$$

$$x = \frac{1}{2} \text{ or } x = 4$$

- (b) The number of words per minute, N , that a trainee typist can type after t weeks of a course is modelled by $N = 40 - Ae^{-kt}$, where A and k are positive constants. At the start of the course the trainee types 10 words per minute, and after 4 weeks types 25 words per minute.

- (i) Find the value of A . [2]

$$40 - Ae^0 = 10$$

$$A = 30$$

- (ii) Find the exact value of k . [3]

$$40 - 30e^{-4k} = 25$$

$$e^{-4k} = \frac{1}{2}$$

$$-4k = \ln \frac{1}{2}$$

$$-4k = -\ln 2$$

$$k = \frac{\ln 2}{4}$$

- (iii) Explain why, according to this model, the trainee's typing speed never reaches 40 words per minute. [2]

For all $t \geq 0$,

$$e^{-kt} > 0$$

$$30e^{-kt} > 0.$$

Hence $N = 40 - 30e^{-kt} < 40$ for every t .

- 4 It is given that $x^2 - 2x - 3$ is a factor of the expression $2x^3 + ax^2 + bx - 3$, where a and b are constants.

(a) Find the value of a and of b . [4]

$$x^2 - 2x - 3 = (x - 3)(x + 1), \text{ so } x = 3 \text{ and } x = -1 \text{ are roots.}$$

$$f(3) = 0$$

$$54 + 9a + 3b - 3 = 0$$

$$3a + b = -17$$

$$f(-1) = 0$$

$$-2 + a - b - 3 = 0$$

$$a - b = 5$$

$$\therefore a = -3, b = -8$$

(b) Find the other factor of the expression. [2]

$$2x^3 - 3x^2 - 8x - 3 = (x^2 - 2x - 3)(2x + 1)$$

$$\text{other factor} = 2x + 1$$

- (c) Hence, or otherwise, solve the equation $3w^3 + 8w^2 + 3w - 2 = 0$. [2]

$$3w^3 + 8w^2 + 3w - 2 = 0$$

$$2 - 3w - 8w^2 - 3w^3 = 0$$

$$2\left(\frac{1}{w^3}\right) - 3\left(\frac{1}{w^2}\right) - 8\left(\frac{1}{w}\right) - 3 = 0$$

$$\therefore \frac{1}{w} = 3, -1, -\frac{1}{2}.$$

$$w = \frac{1}{3}, -1, -2$$

- 5 (a) Given that $\cos A = \frac{4}{5}$ and A is a reflex angle, find the exact value of $\cos 2A$.

$$\begin{aligned}
 \cos 2A &= 2 \cos^2 A - 1 & \cos 2A &= 1 - 2 \sin^2 A & [2] \\
 &= 2 \left(\frac{16}{25} \right) - 1 & \text{or} & & = 1 - 2 \left(-\frac{3}{5} \right)^2 \\
 &= \frac{7}{25} & & & = \frac{7}{25}
 \end{aligned}$$

- (b) Find the exact value of $\cos(A + 60^\circ)$, giving your answer in the form $\frac{a + b\sqrt{3}}{10}$, where a and b are integers. [2]

A reflex with $\cos A > 0 \Rightarrow A$ is in the fourth quadrant, so $\sin A = -\frac{3}{5}$.

$$\begin{aligned}
 \cos(A + 60^\circ) &= \cos A \cos 60^\circ - \sin A \sin 60^\circ \\
 &= \frac{4}{5} \cdot \frac{1}{2} - \left(-\frac{3}{5} \right) \cdot \frac{\sqrt{3}}{2} \\
 &= \frac{4 + 3\sqrt{3}}{10}
 \end{aligned}$$

- (c) Solve the equation $2\cos^2 \theta - 3\sin \theta - 3 = 0$ for $0^\circ \leq \theta \leq 360^\circ$. [4]

$$2(1 - \sin^2 \theta) - 3\sin \theta - 3 = 0$$

$$2\sin^2 \theta + 3\sin \theta + 1 = 0$$

$$(2\sin \theta + 1)(\sin \theta + 1) = 0$$

$$\sin \theta = -\frac{1}{2} \text{ or } \sin \theta = -1$$

$$\alpha = 30^\circ \text{ or } \alpha = 90^\circ$$

$$\theta = 210^\circ, 330^\circ, 270^\circ$$

- 6 (a) The graph of $y = a \cos(bx) + c$, where a , b and c are positive constants, is defined for $0^\circ \leq x \leq 360^\circ$. It has a maximum value of 7, a minimum value of -1 , and a period of 120° . State the value of a , b and c . [3]

$$\text{Amplitude: } a = \frac{7 - (-1)}{2} = 4$$

$$\text{midline: } c = \frac{7 + (-1)}{2} = 3$$

$$\text{period: } \frac{360^\circ}{b} = 120^\circ$$

$$b = 3$$

- (b) It is given that $y = 4 \cos \theta + 5 \sin \theta$.

- (i) Express y in the form $R \sin(\theta + \alpha)$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$. [3]

$$R \sin \alpha = 4, R \cos \alpha = 5$$

$$R = \sqrt{5^2 + 4^2} = \sqrt{41}$$

$$\tan \alpha = \frac{4}{5}$$

$$\alpha = 38.660^\circ$$

$$y = \sqrt{41} \sin(\theta + 38.7^\circ)$$

- (ii) Write down the least value of $7 + 4\cos\theta + 5\sin\theta$. [2]

The least value of $\sqrt{41}\sin(\theta + 38.660^\circ)$ is $-\sqrt{41}$ or -6.40 (3 s.f.).

$\therefore$ least value of $7 + 4\cos\theta + 5\sin\theta = 7 - \sqrt{41}$ or 0.597 (3 s.f.)

- (iii) Find the maximum value of $\frac{1}{7 + 4\cos\theta + 5\sin\theta}$, and the value of θ , for $0^\circ \leq \theta \leq 360^\circ$, at which this maximum occurs. [3]

$\frac{1}{7 + 4\cos\theta + 5\sin\theta}$ is greatest when $7 + 4\cos\theta + 5\sin\theta$ is least,
i.e. $7 - \sqrt{41}$

$$\text{Maximum} = \frac{1}{7 - \sqrt{41}} = \frac{7 + \sqrt{41}}{8} \text{ or } 1.68 \text{ (3 s.f.)}$$

$$\text{Least when } \sin(\theta + 38.660^\circ) = -1$$

$$\theta + 38.660^\circ = 270^\circ$$

$$\theta = 231.3^\circ$$

- (iv) Explain why $\frac{1}{7 + 4\cos\theta + 5\sin\theta}$ is defined for all values of θ . [2]

The least value of $7 + 4\cos\theta + 5\sin\theta$ is $7 - \sqrt{41}$,

and $\sqrt{41} < \sqrt{49} = 7$,

so $7 - \sqrt{41} > 0$.

Hence $7 + 4\cos\theta + 5\sin\theta > 0$ for every θ , it is never zero,
so the fraction is always defined.

7 Two points have coordinates $A(2,4)$ and $B(b,-2)$, where b is a constant with $b > 0$.

- (a) Given that the perpendicular bisector of AB passes through the origin O ,
show that $b = 4$. [4]

$$\text{midpoint of } AB = \left(\frac{2+b}{2}, 1 \right)$$

$$\text{gradient of } AB = \frac{-6}{b-2}$$

$$\text{gradient of perp bisector} = \frac{b-2}{6}.$$

gradient of perp bisector = gradient O to midpoint

$$\frac{b-2}{6} = \frac{2}{2+b}$$

$$(b-2)(b+2) = 12$$

$$b^2 = 16$$

$$b = 4, -4 \text{ (rej.)}$$

- (b) (i) Find the equation of the perpendicular bisector of AB . [3]

$$\text{midpoint} = (3,1)$$

$$\text{perpendicular gradient} = \frac{4-2}{6}$$

$$= \frac{1}{3}$$

$$y-1 = \frac{1}{3}(x-3)$$

$$y = \frac{x}{3}$$

- (ii) The perpendicular bisector of AB meets the line $x - y = 6$ at the point P .
Find the coordinates of P , and show that P is equidistant from A and B .

[3]

$$x - \frac{x}{3} = 6$$

$$x = 9, y = 3$$

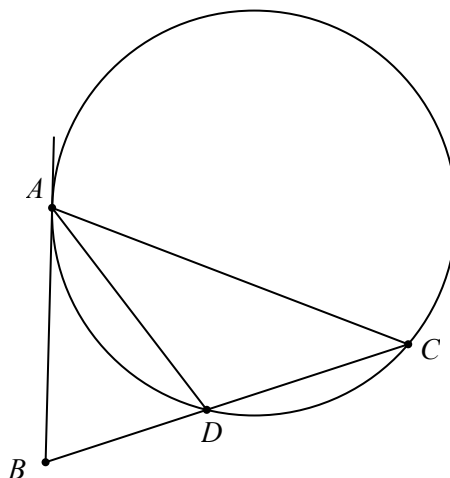
$$\therefore P = (9, 3)$$

$$PA = \sqrt{(9-2)^2 + (3-4)^2} = \sqrt{50}$$

$$PB = \sqrt{(9-4)^2 + (3+2)^2} = \sqrt{50}$$

$$PA = PB = 5\sqrt{2}, \text{ so } P \text{ is equidistant from } A \text{ and } B$$

- 8 In the diagram, B , D and C are points on a straight line and $\angle BAD = \angle ACB$.
A circle passes through the points A , D and C .



- (a) Prove that triangle ABD is similar to triangle CBA . [2]

$$\angle ABD = \angle CBA \text{ (common } \angle \text{)}$$

$$\angle BAD = \angle BCA \text{ (given)}$$

$\therefore$ triangle ABD is similar to triangle CBA (AA similarity)

- (b) Given that $BD = 4$ cm and $DC = 5$ cm, find the length of BA . [3]

$$\frac{BD}{BA} = \frac{BA}{BC}$$

$$\frac{4}{BA} = \frac{BA}{4+5}$$

$$BA^2 = 4 \times 9$$

$$BA = 6 \text{ cm}$$

- (c) Prove that BA is a tangent to the circle at A . [2]

Since it is given that $\angle BAD = \angle ACB$,
by the converse of the alternate segment theorem, BA is a tangent to the circle at A .

- 9 The equation of a curve is $y = 2x^3 - 3(k+2)x^2 + 12kx$, where k is a constant and $k > 2$.

(a) Show that the coordinates of the two stationary points of the curve are $(2, 12k - 8)$ and $(k, -k^3 + 6k^2)$.

[5]

$$\frac{dy}{dx} = 0$$

$$6x^2 - 6(k+2)x + 12k = 0$$

$$6(x-2)(x-k) = 0$$

$$x = 2 \text{ or } x = k$$

For $x = 2$,

$$y = 16 - 12(k+2) + 24k$$

$$= 12k - 8$$

For $x = k$,

$$y = 2k^3 - 3(k+2)k^2 + 12k^2$$

$$= -k^3 + 6k^2$$

$\therefore$ Stationary points: $(2, 12k - 8)$ and $(k, -k^3 + 6k^2)$.

- (b) Given that the difference between the maximum value and the minimum value of y is 27, find the value of k . [5]

Since the leading coefficient is positive and $k > 2$,

$x = 2$ gives the maximum and $x = k$ the minimum

$$(12k - 8) - (-k^3 + 6k^2) = 27$$

$$k^3 - 6k^2 + 12k - 8 = 27$$

$$k^3 - 6k^2 + 12k - 35 = 0$$

For $f(k) = k^3 - 6k^2 + 12k - 35$, when $k = 5$,

$$f(k) = 0$$

$$(k - 5)(k^2 - k + 7) = 0$$

$\therefore k = 5$ or $k^2 - k + 7 = 0$ (rej. with quad formula or discriminant < 0)

- (c) Determine, justifying your answer, the range of values of k for which the minimum value of y is negative. [3]

When $x = k$,

$$y = -k^3 + 6k^2$$

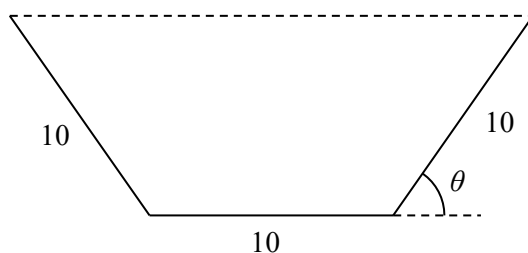
$$= k^2(6 - k)$$

Since $k > 2$, $k^2 > 0$,

so $6 - k < 0$

$\therefore k > 6$

- 10** A water channel is made by bending a long rectangular strip of metal, of width 30 cm, into three equal sections each of width 10 cm, as shown in the cross-section. The middle section forms a horizontal base and each of the two outer sections is turned up through an angle θ , where $0^\circ < \theta < 90^\circ$. The cross-sectional area of the channel is $A \text{ cm}^2$.



- (a)** Show that $A = 100 \sin \theta + 50 \sin 2\theta$. [2]

$$\begin{aligned} A &= \frac{1}{2} (10 + 10 + 20 \cos \theta) (10 \sin \theta) \\ &= 100 \sin \theta + 100 \sin \theta \cos \theta \\ &= 100 \sin \theta + 50 \sin 2\theta \end{aligned}$$

- (b)** Given that θ can vary, find the value of θ for which A has a stationary value. [4]

$$\begin{aligned} \frac{dA}{d\theta} &= 0 \\ 100 \cos \theta + 100 \cos 2\theta &= 0 \\ \cos \theta + (2 \cos^2 \theta - 1) &= 0 \\ 2 \cos^2 \theta + \cos \theta - 1 &= 0 \\ (2 \cos \theta - 1)(\cos \theta + 1) &= 0 \\ \cos \theta &= \frac{1}{2} \text{ (rejecting } \cos \theta = -1), \text{ so } \theta = 60^\circ. \end{aligned}$$

- (c) Find the maximum cross-sectional area and verify that it is a maximum. [2]

$$A = 100 \sin 60^\circ + 50 \sin 120^\circ$$

$$= 100 \left(\frac{\sqrt{3}}{2} \right) + 50 \left(\frac{\sqrt{3}}{2} \right)$$

$$= 75\sqrt{3} \text{ cm}^2.$$

$$\frac{d^2 A}{d\theta^2} = -100 \sin \theta - 200 \sin 2\theta$$

$$= -100 \sin 60^\circ - 200 \sin 120^\circ < 0, \text{ so it is a maximum.}$$

END OF PAPER

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